Substance Use Disorders in Adolescents

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Introduction

Substance use in adolescence can have widespread negative effects on adolescent health, including school failure, respiratory disease, high-risk sexual behaviors, injury, and even death. Although some view the occasional use of substances such as alcohol or marijuana as part of normal adolescent development, there exists significant potential for injury even with occasional use.

Adolescents with substance use problems present in a variety of ways:

- ☑ Intoxicated in the emergency department
- ☑ In your clinic with physical complaints, e.g., persistent abdominal pain, palpitations, chest pain, recurrent sore throat, and persistent nasal discharge
- ☑ Brought in by parents for behavioral concerns, changes in school performance, sudden mood swings, stealing, or lying
- ☑ No presenting signs or symptoms; disclosed only on direct questioning by healthcare professionals

Substance use patterns in adolescents are presented in **Figure 1**.

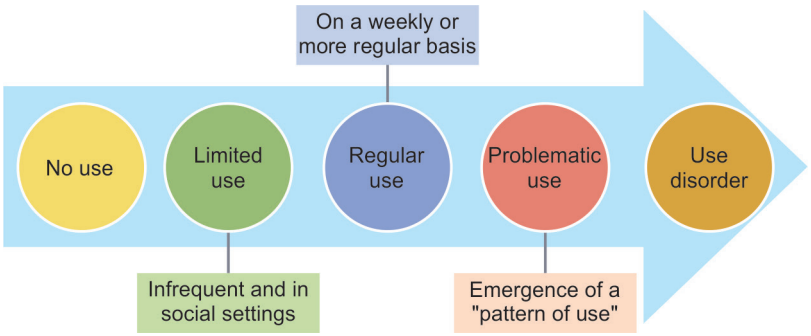


Fig. 1: Substance use patterns in adolescents.

In Emergency Room

General Management

Intoxicated adolescents should be assessed and managed through the ABCDE approach. Early resuscitation is a life-saving measure in intoxicated patients. A—Maintain Airway and give Antidotes, B—Facilitate Breathing, C—Circulation, D—Manage Disability, and E—Avoid Exposure to dangers.

Management of Specific Substances (Over and above Symptomatic Therapy)

☑ Find out drug-specific indicators during history taking and physical examination:

- *Amnesia*: Barbiturates, ketamine, and flunitrazepam
- *Gastritis, blackouts, and body odor*: Alcohol
- *Flu-like symptoms*: Cocaine, inhalants, and marijuana
- *Red conjunctiva and abnormal pupils*: Marijuana
- *Paint stains on face and clothes*: Inhalants
- *Chest pain and tachycardia*: Cocaine and amphetamine
- *Bronchospasm and constipation*: Opiates and marijuana
- *Increased appetite for sweets*: Marijuana
- *Altered body odor*: Alcohol and inhalants
- *Gynecomastia, small testes, and irregular periods*: Marijuana
- *Gasoline cans in car*: Inhalants

☑ Specific management:

- *Alcohol*: Intravenous (IV) fluids with glucose and vitamins
- *Cannabis*: Rarely serious and may not require anything else than the symptomatic treatment including benzodiazepines
- *Opioids*: IV naloxone-bolus 1–2 mg followed by infusion at 0.1 mg/kg/hour, maximum up to 10 mg total
- *Stimulants*: Benzodiazepines, aggressive cooling, and sodium bicarbonate for cocaine
- *Benzodiazepine*: Flumazenil can be used in severe intoxication associated with respiratory and neurological depression.

In Nonemergency Settings

Management of substance use in adolescents needs team work from multiple specialties. Pediatricians are the ones who should screen them and guide them during routine office visits. HEEADSSS method of psychosocial assessment should be done in all adolescents visiting pediatric office to screen for substance use disorder (SUD). Look for red flag signs, i.e., headache, sore throat, worsening asthma, chronic cough, chest pain, gastritis, hepatitis, needle puncture marks, and pancreatitis.

- ☑ *Alcohol*: Disulfiram (inhibits an enzyme involved in the metabolism of alcohol, causing an unpleasant reaction if alcohol is consumed after taking the medication), naltrexone (decreases alcohol-induced euphoria), acamprosate (reduces withdrawal symptoms), ondansetron, selective serotonin reuptake inhibitor (SSRI), topiramate, and baclofen
- ☑ *Opioid*: Replacement therapy: buprenorphine-naloxone and methadone (all act by both activating and blocking opioid receptors in the brain
- ☑ *Cannabis*: N-acetylcysteine and baclofen
- ☑ *Inhalants*: Baclofen
- ☑ *Nicotine*: Bupropion (reduces nicotine cravings and withdrawal symptoms and depression), varenicline (reduces nicotine cravings and withdrawal symptoms by stimulating nicotine receptors in the brain), nicotine replacement therapies: activate nicotine receptors in the brain, available in the form of a patch, gum, lozenge, nasal spray, and inhaler.

CRAFFT questionnaire should be used to assist Screening to Brief Intervention (S2BI) method to screen adolescents for SUD **(Fig. 2)**.

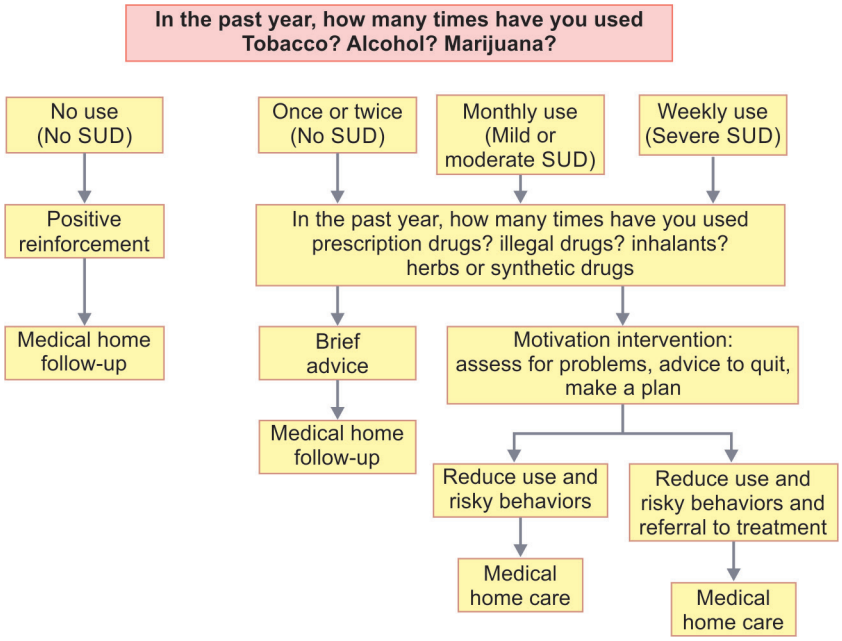
S2BI: Screening to brief intervention In the past year, how many times have you used	
• Tobacco? • Alcohol? • Marijuana? Stop if all "Never". Otherwise, continue • Prescription drugs that were not prescribed for you (such as pain medication or adderall)? • Illegal drugs (such as cocaine or Ecstasy)? • Inhalants (such as nitrous oxide)? • Herbs or synthetic drugs (such as salvia, "K2", or bath salts)?	◯ Never ◯ Once or twice ◯ Monthly ◯ Weekly

Fig. 2: Screening to brief intervention.

CRAFFT Screening Questionnaire

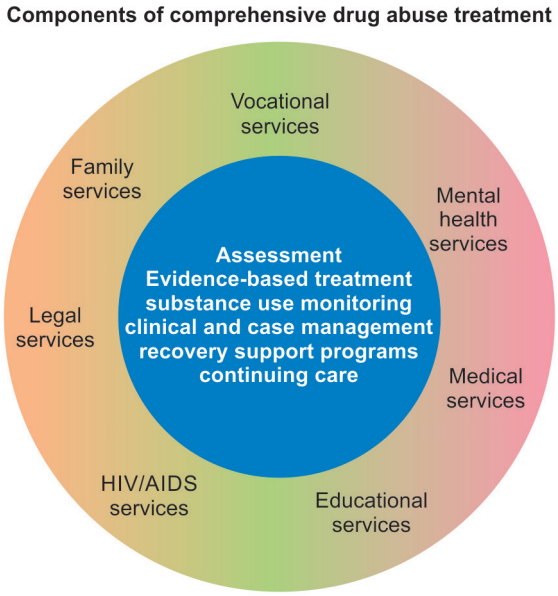
C: Have you ever ridden in a **C**ar driven by someone who was high or had been using drugs?
R: Do you ever use drugs to **R**elax or feel better or to fit in?
A: Do you ever use drugs when you are **A**lone?
F: Do you ever **F**orget things when you are on drugs?
F: Do your **F**amily/**F**riends ask you to cut down on drugs?
T: Have you ever gotten into **T**rouble while using drugs?

Flowchart 1: Algorithm for management of substance use disorder in adolescence.



(SUD: substance use disorder)

(*Source:* Levy S, Shrier L. The S2BI-based approach to clinical SBIRT. Boston, MA: Boston Children's Hospital; 2014.)



The best treatment programs provide a combination of therapies

Fig. 3: Components of comprehensive drug abuse treatment.